Final Project

PSTAT122: Design and Analysis of Experiments, Spring 2026

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1 Introduction

When a visual scene suddenly changes, our reaction speed depends in part on how big the change in the scene is. A small modification in shade can be brushed off as margin of error in our vision, but a shoft to the polar opposite possibility cant be missed. This is the intuition behind many of the worlds most critical safety systems, and it is the basis of what we decided to focus this project on

This project asks a focused version of that question:

> **Project question:** Does the magnitude of a color change affect how quickly a person reacts to it?

We built a web app in which the screen starts black and, after a random delay, jumps to one of eight shades of gray. The participant reacts as soon as they detect the change. Because every trial starts from the same black baseline, the gray value of the target (0–255 RGB) is the magnitude of the change — a small value like 32 is a subtle dark-gray, while 255 is a full black-to-white jump. Our hypothesis is that larger changes produce faster reactions, and that the effect is part of a gradient rather than all-or-none. Because people differ in baseline reaction speed, we used RCBD, blocking on participant so that each person serves as their own control, and only the differences are measured from that.

2 Experimental Design

Response variable. Simple visual reaction time, in milliseconds. It is the elapsed time from the time the screen changes color to the time the participant hits the space bar

Treatments The magnitude of the color change, the target gray level on the 0–255 RGB scale. We used 8 equally spaced levels: 32, 64, 96, 128, 160, 192, 224, 255. Since every trial begins at pure black (0), each level is exactly the size of the luminance jump. This factor is fixed — we chose these evenly spaced magnitudes and wanted to fine the response relationship across them. The equal spacing is what later lets us decompose the treatment effect into interpretable trend components.

Blocking The participants (Tejas, Nathan, Josh) — 3 blocks. Blocking on person removes the variation of baseline reaction speed by comparing the eight contrast levels within each individual. We treat the block as a random effect. For the additive RCBD, the treatment F-test is identical whether blocks are regarded as fixed or random.

Design type. RCBD with replication. Every block (participant) is exposed to all 8 treatments, so the blocks are complete. The additive model is

$$\log(\mathrm{RT}_{ijk}) \;=\; \mu \;+\; \tau_i \;+\; \beta_j \;+\; \varepsilon_{ijk}, \qquad i = 1, \ldots, 8;\; j = 1, 2, 3,$$

where τ_i is the (fixed) effect of contrast level i, β_j the block effect of participant j, and ε_{ijk} the within-cell error. (We model RT on the log scale; see Section 4.)

Randomization, replication, and blocking.

- Randomization: Within every session the app uses a Fisher–Yates shuffle to randomize the order of the 40 color presentations independently each time. This breaks any confound between contrast level and trial position, so learning, warm-up, and fatigue cannot bias the comparison among levels.
- Replication: Each participant completed 7 sessions of 40 trials, with each gray level shown 5 times per session, giving 35 replicate measurements per participant-by-level cell.
- Blocking: Participant is the blocking variable. Each block is internally homogeneous while large differences may exist between blocks.

Sample size. With 35 replicates per cell, 105 observations per contrast level, and 840 trials in total, the design far exceeds the textbook RCBD guideline. The heavy within-block replication is intentional since single reaction times are extremely noisy and milliseconds are a very precise time unit (SDs near 80–130 ms), so many repeats per cell are needed to estimate each cell mean precisely. Each session lasts about three minutes, so the entire data-collection burden per person is only about 25 minutes, well under an hour.

3 Data Collection

Procedure: Data was collected between 29 and 31 May 2026 using the browser app. On each visit a participant entered their name, read the instructions, and completed one unlogged warm-up trial followed by 40 logged trials. Every trial proceeded identically: the screen showed black, and after a random delay of 1–4 seconds (uniformly drawn to prevent anticipation) it changed to the trial's target color. The participant would hit the space bar as fast as they could; the site recorded the gray level and the reaction time and automatically advanced to the next trial. Each participant repeated the session 7 times on separate occasions. Results were never shown to the participant during the test to prevent them from trying to outperform themselves between trials, and each completed trial was streamed to an excel file via the Google Apps Script API.

Challenges and adjustments.

- Anticipation deterrence. If a participant responded before the color changed, the app flagged the attempt as a jump and silently re-ran that trial. Premature guesses were not logged as valid responses.
- Hidden scores. Reaction times were deliberately concealed from participants during a session to discourage gaming or score-chasing, which would otherwise distort natural responses.
- Uncontrolled hardware and environment. Participants used their own laptops in their own lighting, so display brightness, refresh rate, and input latency varied between people and between sessions. This adds measurement noise; the design absorbs it into the block (participant) and session structure rather than into the treatment comparison.
- Invalid responses. As in any reaction time study, a handful of responses were anticipations (impossibly fast) or lapses of attention (very slow). These are handled transparently in Section 4, with responses above +3 stdev and below 150ms being cut out, as that is faster than the time it takes for an optical signal to go from your eye to your brain.

Data presentation: The cleaned data contain 822 of the 840 recorded reaction times. Table 1 summarizes RT by contrast level and Table 2 (both below) by participant. The distributions are explored visually with graphs in the Analysis section.

Table 1: Reaction time (ms) by contrast level (cleaned data). Means decline as the black-to-gray change grows larger.

Gray level	n	Mean (ms)	SD (ms)	Median (ms)
32	103	354.2	87.7	331.0
64	99	338.0	113.5	308.0
96	105	329.0	87.0	298.0
128	102	326.5	92.9	301.0
160	103	333.4	117.2	297.0

Table 1: Reaction time (ms) by contrast level (cleaned data). Means decline as the black-to-gray change grows larger.

Gray level	n	Mean (ms)	SD (ms)	Median (ms)
192	105	318.4	95.0	299.0
224	101	321.0	101.2	299.0
255	104	314.6	89.4	287.5

Table 2: Reaction time (ms) by participant. Large between-person differences in speed and variability make blocking worth it.

Participant	n	Mean (ms)	SD (ms)	Median (ms)
Josh	274	296.8	75.9	276.0
Nathan	274	376.5	127.4	336.5
Tejas	274	314.7	61.9	297.0

4 Analysis

Data cleaning: Following standard practice for reaction-time data, we removed two classes of invalid trials before analysis: anticipations — responses faster than 150 ms, which is below the physical floor for visual RT (we found 2) — and lapses — responses slower than three standard deviations above a participant's own mean (16 trials). This removed 18 of 840 trials (2.1%), leaving 822 (97.9%) for analysis; the design remained essentially balanced all with 33+ replicates per cell. All conclusions below are robust to these choices.

Exploratory analysis: Mean reaction time declined steadily with contrast magnitude, from 354 ms at the darkest level (32) to 315 ms at the brightest (255) — a drop of about 40 ms, or 11.2%. Participants differed much more: mean reaction times ranged from 297 to 377 ms, and the slowest participant was also the most variable. Crucially, the downward trend with contrast appears in all three participants and the per-person curves are roughly parallel, which means the effects are additive.

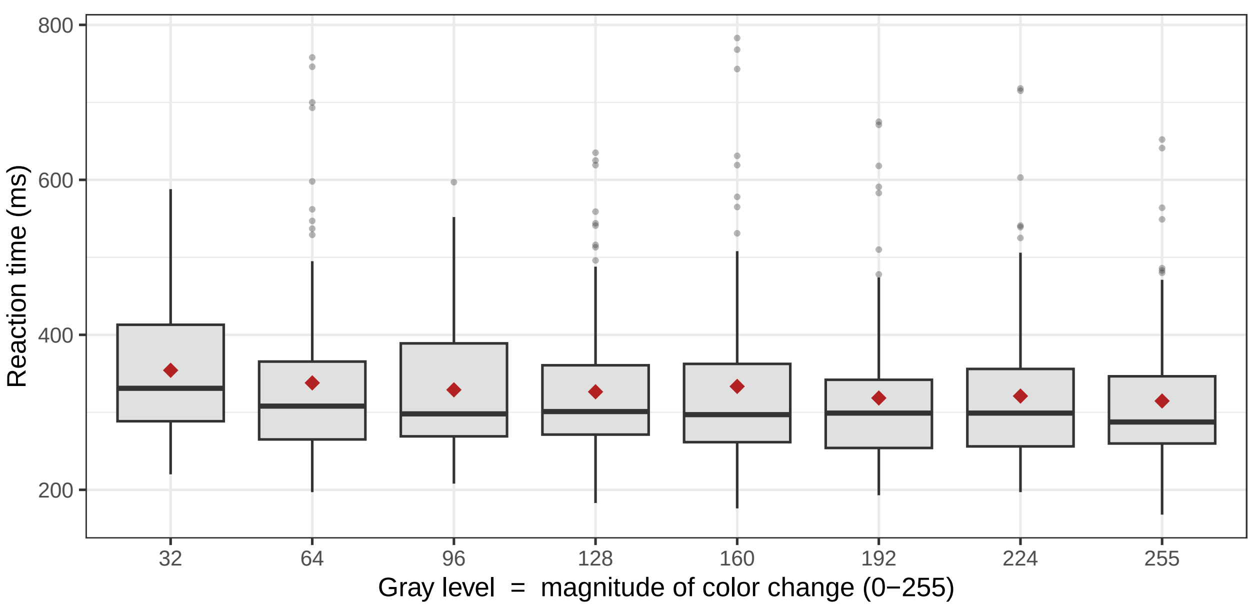

Figure 1: Reaction time by contrast level (gray value). Boxes show median and IQR, red diamonds are means. Reaction time decreases gradually as the magnitude of the black-to-gray change increases.

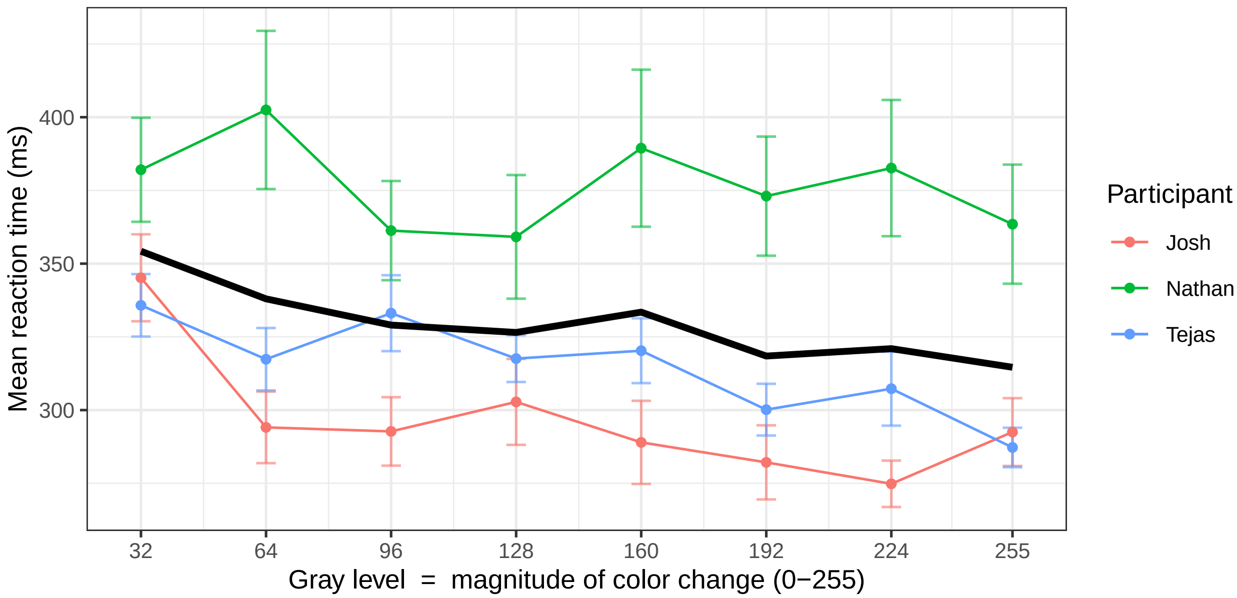

Figure 2: Mean reaction time +/-1 SE vs contrast magnitude for each participant (colored) and overall (black).

We fit the RCBD model with contrast level (the 8-level treatment) and participant (the blocks). Reaction times are strongly right-skewed, so the raw-scale model most definitely violated normality (Shapiro–Wilk $p < 0.001$). Modeling log(RT) resolved this: residuals were approximately normal (Shapiro–Wilk $W = 0.97$) and homogeneous across levels. That's why we based our inference on the log scale and reported effects back-transformed to a percentage scale. The block × treatment interaction was not significant F(14,798)=0.96 and p=0.498, so the additive RCBD is appropriate.

Hypothesis test: We test $H_0 : \tau_1 = \cdots = \tau_8$ (contrast magnitude has no effect on mean log-RT) against the alternative that at least one of the means differ significantly from the test. The ANOVA Table rejects the null hypothesis, meaning the contrast level is significant, F(7,812)=2.57 and p=0.013. The block effect is very large (F is 54 and $p < 0.001$), confirming that assumption that participants differ. This also shows that blocking was worth doing.

Because the eight gray levels are evenly spaced, the most powerful way to test our hypothesis is to ask whether reaction time follows a steady trend as the change gets bigger. We split the overall treatment effect into a straight-line (linear) trend plus curved, higher-order trends — a standard breakdown known as orthogonal polynomial contrasts. The straight-line trend is highly significant ($F(1, 812) = 14.6$, $p < 0.001$) and explains almost all of the difference among the eight levels, while the quadratic trend ($p = 0.243$) and all higher-order terms are not significant. In plain terms, reaction time drops in a roughly straight line as the color change grows larger. If we treat the gray level as a plain number and fit a simple regression, each one-unit increase lowers reaction time by about 0.05%, which adds up to about a 9.6% speed-up across the whole range (32 to 255).

Table 3: RCBD ANOVA on log(reaction time), with the treatment effect decomposed into trend components.

Source	Df	Sum Sq	Mean Sq	F	p
Contrast level (treatment)	7	1.10	0.16	2.57	0.013
Linear trend	1	0.89	0.89	14.56	< 0.001
Quadratic trend	1	0.08	0.08	1.36	0.243
Participant (block)	2	6.63	3.31	54.17	< 0.001
Residuals	812	49.68	0.06		

Post-hoc comparisons: We used Bonferroni's Correction as our post-hoc comparison. Pairwise comparisons across all 28 pairs of contrast levels found 3 significant at the 0.05 level: the darkest level of 32 was significantly slower than each of the three highest-contrast levels (192, 224, 255). No two adjacent levels differed significantly, which is what our hypothesis predicted.

Table 4: Significant pairwise comparisons on a log scale. Only 32 differs significantly from the others. Change is difference in mean reaction time

Comparison	Diff (log)	Change	Adj. p
32 vs 255	0.126	+13.4%	0.0077
32 vs 192	0.115	+12.2%	0.0241
32 vs 224	0.109	+11.6%	0.0457

Was blocking worth it? Yes. The estimated relative efficiency of this RCBD vs a completely randomized design is about 5.8, meaning if we did a CRD we would have needed about 6x times as many observations to match the precision we obtained by blocking on participant. This large gain reflects the wide between-person spread seen in the line graph.

Robustness: The contrast effect does not depend on our analysis choices. It remains significant p=0.03 with no trimming at all and with only anticipations removed $p = 0.009$; the linear trend is significant on the raw scale as well as the log scale. Most importantly, because trials are repeated within sessions, we refit the model as a mixed-effects model with participant and session as random effects, properly accounting for the clustering instead of counting all trials as independent. The contrast effect survives ($\chi^2(7) = 29.2$; $p < 0.001$). That also reveals where the variability in the results comes from. The session-to-session fluctuation within a person (SD 0.16 on the log scale) is larger than the difference between each participant (0.07), and trial-to-trial noise being 0.20 SD.

5 Conclusions

Key findings: The magnitude of a color change has a small but statistically significant effect on reaction time. Larger color contrasts are detected faster, and the relationship is essentially linear over the range. Reaction time falls by roughly 40 ms or 11.2% from the 32 to 255. The effect holds across every participant, shows no block $\times$ treatment interaction, and persists under a mixed-effects model that accounts for repeated sessions. This is consistent with the expectation that stronger stimuli are processed faster.

At the same time, the relative effect size also matters: contrast magnitude explains only about 1.9% of the variance in log-RT, while individual differences account for about 11.5% and ordinary trial-to-trial noise contributing the remaining 86.5%. So while bigger changes really are noticed faster, the person reacting and random noise matter far more than the size of the change. There is a difference but you hit siminishing returns pretty fast afterwards

Limitations: (1) Only three participants, all of similar age, gender, and background, were measured, so generalization to global population is not applicable. (2) Computer and display, ambient lighting and input latency were not accounted for, which likely inflated measurement noise. (3) The baseline was always black, so we measured detection of luminance increments of varying size — not arbitrary differences or decrements.

Possible improvements: Recruit more participants sampled from a defined population, make everyone use the same screen with the same brightness, record time-of-day, extend the treatment factor to include going from bright to dark and also other non-grayscale colors, and given the graded response, model RT with a nonlinear function rather than a linear trend.

6 Appendix

All R code used to load and clean the data, fit the models, and produce every table and figure in this report is collected below.

```
knitr::opts_chunk$set(message = FALSE, warning = FALSE, error = FALSE, echo = FALSE)
## ---- Packages ----
library(readr); library(dplyr); library(tidyr); library(stringr)
library(ggplot2); library(forcats); library(car); library(emmeans)
library(lme4); library(broom); library(knitr)

theme_set(theme_bw(base_size = 10))

f2 <- function(x) formatC(x, format = "f", digits = 2)

raw <- read_csv("Data.csv", show_col_types = FALSE)
names(raw) <- c("name_session", "trial", "gray", "rt")
dat <- raw |>
  mutate(
    subject = word(name_session, 1),                   # "Josh 3" -> "Josh"
    session = as.integer(word(name_session, -1)),      # "Josh 3" -> 3
    gray    = as.integer(gray),
    rt      = as.numeric(rt)
  )
N_total   <- nrow(dat)
n_subj    <- n_distinct(dat$subject)
n_sess    <- n_distinct(dat$session)
n_levels  <- n_distinct(dat$gray)
gray_vals <- sort(unique(dat$gray))
reps_cell <- N_total / (n_subj * n_levels)             # replicates per subject x level

N_anti <- sum(dat$rt < 150)
fence <- dat |>
  filter(rt >= 150) |>
  group_by(subject) |>
  summarise(hi = mean(rt) + 3 * sd(rt), .groups = "drop")
clean <- dat |>
  filter(rt >= 150) |>
  left_join(fence, by = "subject") |>
  filter(rt <= hi) |>
  transmute(
    subject = factor(subject),
    session,
    trial,
    gray,
    grayF   = factor(gray, levels = gray_vals),
    rt,
    log_rt  = log(rt)
  )
N_clean   <- nrow(clean)
N_removed <- N_total - N_clean
N_lapse   <- N_removed - N_anti
pct_kept  <- 100 * N_clean / N_total
```

```r
mean_dark  <- mean(clean$rt[clean$gray == 32])
mean_light <- mean(clean$rt[clean$gray == 255])
diff_dl    <- mean_dark - mean_light
pct_dl     <- 100 * diff_dl / mean_dark

m_raw <- aov(rt ~ grayF + subject, data = clean)
m_log <- aov(log_rt ~ grayF + subject, data = clean)
a_log <- anova(lm(log_rt ~ grayF + subject, data = clean))
df_res  <- a_log["Residuals", "Df"]
F_gray  <- a_log["grayF", "F value"];   p_gray  <- a_log["grayF", "Pr(>F)"]
F_subj  <- a_log["subject", "F value"]; p_subj  <- a_log["subject", "Pr(>F)"]

sh_raw  <- shapiro.test(residuals(m_raw))$p.value
sh_logW <- shapiro.test(residuals(m_log))$statistic
lev_log <- leveneTest(log_rt ~ grayF, data = clean)[1, "Pr(>F)"]

a_int  <- anova(lm(log_rt ~ grayF * subject, data = clean))
F_int  <- a_int["grayF:subject", "F value"]
p_int  <- a_int["grayF:subject", "Pr(>F)"]
df_i1  <- a_int["grayF:subject", "Df"]
df_i2  <- a_int["Residuals", "Df"]


clean_poly <- clean
contrasts(clean_poly$grayF) <- contr.poly(n_levels)
m_poly <- aov(log_rt ~ grayF + subject, data = clean_poly)
sp  <- summary(m_poly, split = list(grayF = list(Linear = 1, Quadratic = 2)))[[1]]
spn <- trimws(rownames(sp))
grab <- function(nm, col) as.numeric(sp[which(spn == nm), col])
F_lin  <- grab("grayF: Linear", 4);    p_lin  <- grab("grayF: Linear", 5)
F_quad <- grab("grayF: Quadratic", 4); p_quad <- grab("grayF: Quadratic", 5)
SS_gray <- grab("grayF", 2);            MS_gray <- grab("grayF", 3)
SS_lin  <- grab("grayF: Linear", 2);    MS_lin  <- grab("grayF: Linear", 3)
SS_quad <- grab("grayF: Quadratic", 2); MS_quad <- grab("grayF: Quadratic", 3)
SS_subj <- grab("subject", 2);          MS_subj <- grab("subject", 3)
SS_res  <- grab("Residuals", 2);        MS_res  <- grab("Residuals", 3)

m_poly_raw <- aov(rt ~ grayF + subject, data = clean_poly)
spr  <- summary(m_poly_raw, split = list(grayF = list(Linear = 1)))[[1]]
sprn <- trimws(rownames(spr))
F_lin_raw <- as.numeric(spr[which(sprn == "grayF: Linear"), 4])
p_lin_raw <- as.numeric(spr[which(sprn == "grayF: Linear"), 5])

m_num    <- lm(log_rt ~ gray + subject, data = clean)
slope    <- unname(coef(m_num)["gray"])
pct_range <- (exp(slope * (255 - 32)) - 1) * 100      # darkest -> lightest, %

emm_gray <- emmeans(m_log, "grayF")
bonf <- as.data.frame(pairs(emm_gray, adjust = "bonferroni"))
bonf$pair <- gsub("grayF", "", gsub(" - ", "-", bonf$contrast))
n_pairs <- nrow(bonf)
n_sig   <- sum(bonf$p.value < 0.05)
```

```r
ar  <- anova(lm(rt ~ grayF + subject, data = clean))
MSB <- ar["subject", "Mean Sq"]; MSE <- ar["Residuals", "Mean Sq"]
b   <- n_subj; t <- n_levels
RE  <- ((b - 1) * MSB + b * (t - 1) * MSE) / ((b * t - 1) * MSE)

clean$sess_id <- interaction(clean$subject, clean$session, drop = TRUE)
mm_full <- lmer(log_rt ~ grayF + (1 | subject) + (1 | sess_id), data = clean, REML = FALSE)
mm_null <- lmer(log_rt ~ 1     + (1 | subject) + (1 | sess_id), data = clean, REML = FALSE)
mm_lrt  <- anova(mm_null, mm_full)
mm_chi  <- mm_lrt$Chisq[2]; mm_df <- mm_lrt$Df[2]; mm_p <- mm_lrt$`Pr(>Chisq)`[2]
vc <- as.data.frame(VarCorr(mm_full))
sd_sess  <- vc$sdcor[vc$grp == "sess_id"]
sd_subj  <- vc$sdcor[vc$grp == "subject"]
sd_resid <- vc$sdcor[vc$grp == "Residual"]

eta       <- a_log[, "Sum Sq"] / sum(a_log[, "Sum Sq"])
eta_gray  <- eta[1]; eta_subj <- eta[2]; eta_resid <- eta[3]
p_none <- anova(lm(log(rt) ~ factor(gray) + factor(subject), data = dat))["factor(gray)", "Pr(>F)"]
p_anti <- anova(lm(log(rt) ~ factor(gray) + factor(subject),
                   data = filter(dat, rt >= 150)))["factor(gray)", "Pr(>F)"]
tab_gray <- clean |>
  group_by(`Gray level` = gray) |>
  summarise(n = n(), `Mean (ms)` = mean(rt), `SD (ms)` = sd(rt),
            `Median (ms)` = median(rt), .groups = "drop")
kable(tab_gray, digits = c(0, 0, 1, 1, 1), booktabs = TRUE, linesep = "")
tab_subj <- clean |>
  group_by(Participant = subject) |>
  summarise(n = n(), `Mean (ms)` = mean(rt), `SD (ms)` = sd(rt),
            `Median (ms)` = median(rt), .groups = "drop")
kable(tab_subj, digits = c(0, 0, 1, 1, 1), booktabs = TRUE, linesep = "")
ggplot(clean, aes(grayF, rt)) +
  geom_boxplot(outlier.size = 0.5, outlier.alpha = 0.35, fill = "grey88") +
  stat_summary(fun = mean, geom = "point", shape = 18, size = 2.4, colour = "firebrick") +
  labs(x = "Gray level  =  magnitude of color change (0-255)", y = "Reaction time (ms)")
summ <- clean |>
  group_by(subject, gray) |>
  summarise(m = mean(rt), se = sd(rt) / sqrt(n()), .groups = "drop")
overall <- clean |>
  group_by(gray) |>
  summarise(m = mean(rt), .groups = "drop")
ggplot(summ, aes(gray, m, colour = subject)) +
  geom_errorbar(aes(ymin = m - se, ymax = m + se), width = 5, alpha = 0.6) +
  geom_line() + geom_point(size = 1.3) +
  geom_line(data = overall, aes(gray, m), colour = "black", linewidth = 1.2, inherit.aes = FALSE) +
  scale_x_continuous(breaks = gray_vals) +
  labs(x = "Gray level  =  magnitude of color change (0-255)",
       y = "Mean reaction time (ms)", colour = "Participant")
F_vec <- c(F_gray, F_lin, F_quad, F_subj, NA)
p_vec <- c(p_gray, p_lin, p_quad, p_subj, NA)
anova_tbl <- data.frame(
  Source = c("Contrast level (treatment)", "\\quad Linear trend", "\\quad Quadratic trend",
             "Participant (block)", "Residuals"),
  Df       = c(7, 1, 1, 2, df_res),
```

```r
  `Sum Sq` = f2(c(SS_gray, SS_lin, SS_quad, SS_subj, SS_res)),
  `Mean Sq`= f2(c(MS_gray, MS_lin, MS_quad, MS_subj, MS_res)),
  `F`      = ifelse(is.na(F_vec), "", f2(F_vec)),
  `p`      = ifelse(is.na(p_vec), "", ifelse(p_vec < 0.001, "< 0.001",
                                           formatC(p_vec, format = "f", digits = 3))),
  check.names = FALSE
)
kable(anova_tbl, booktabs = TRUE, linesep = "", align = "lrrrrr", escape = FALSE)
bonf_sig <- bonf |>
  filter(p.value < 0.05) |>
  arrange(p.value) |>
  transmute(
    Comparison   = gsub("-", " vs ", pair),
    `Diff (log)` = estimate,
    `Change`     = sprintf("%+.1f%%", (exp(estimate) - 1) * 100),
    `Adj. p`     = p.value
  )
kable(bonf_sig, digits = c(0, 3, 0, 4), booktabs = TRUE, linesep = "")
```